

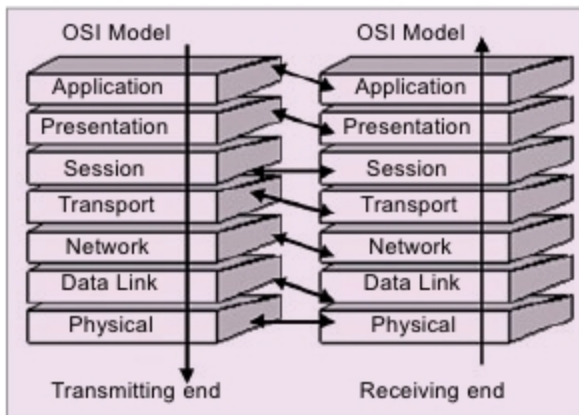


Fig. 2: Illustration of OSI/ISO reference model

sonal secretary follows a prescribed functional protocol to work with the boss. And, an attendant follows a different protocol so that there is no clash in running the office.

In 1984, International Standards' Organization (ISO) of Europe developed an open systems interconnection (OSI) model for communication in networks. OSI is a layered reference framework of standards for communication among different parts of a network and/or among networks made of different equipment and applications by different vendors.

Incompatibilities are resolved by the application of reference model OSI. Over the years, OSI has been accepted as the basic and main architectural model for inter-computing, networking and internetworking communications. As of today, most networks are based on OSI model. Seven layers of OSI reference model divide communication into seven peer processes, which are independent to each other.

The independent nature of implementing functions in each layer makes the design flexible and modular. The function of a layer may be set without affecting the functions of the other layers. For implementing the function of each layer, a set of rules called protocol is defined. In OSI model, within each layer, one or more entities implement their functionality. Each entity interacts directly only with the layer immediately below it on the transmission end and provides facilities for use by the layer above it. Conceptually, at each layer at the transmitting end, a header is added with payload; and at the receiving end, a header is removed, as illustrated in Fig. 2.

Functions of the seven layers of OSI model are:

- The physical layer defines the physical properties (for example, characteristics of signals and codes used for signals) of the various communication media used between a transmitter and a receiver. Physical properties include electrical properties, type of cable used, type of BNC connector used, termination method, signal coding and so on. Cables may be coaxial, Ethernet, twisted pair or other. The layer therefore takes a decision on the method of transmission of bits as physical signals. The logical bit pipe received from the data link layer is converted into physical signals at the transmitter end. Physical signals received by the receiver are converted back into a logical pipe at the receiver end. This makes layer-to-layer functionality a peer process.

- The datalink layer does the logical frame of data bits transmitted/received over the physical layer. At the transmitting end, the layer generates a logical bit pipe with several overheads like address of source and destination, error check sum, length identification and so on, and at the receiving end, the bit pipe is decomposed (by removing the overheads after due processing/checking as the case may be) into received data bits.

Several protocols are used for the organisation of data bits at this layer. Examples of the protocols are SDLC and HDLC. Data bits are put into a frame that includes headers of address (source and destination), control, check sum, and start and end of frame bytes.

LAN is functionally made up of datalink and physical layers. LAN being a shared media network, the datalink layer for LAN has two functions: logical link control (LLC) sub-function and media access control (MAC) sub-function.

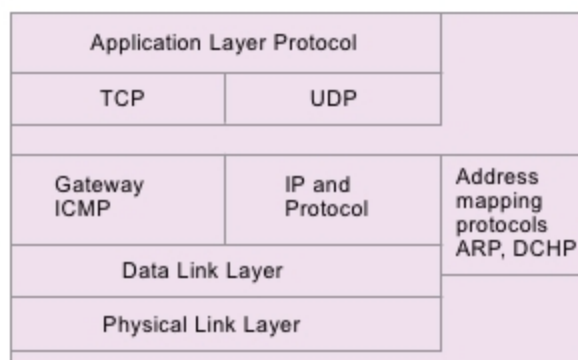


Fig. 3: TCP/IP or UDP/IP suite (Internet) model

- The network layer performs routing and switching. It describes how data packets/frames traverse over various data links in the network between the source and destination. The layer defines the addressing and routing of the Internet. A node is made of three layers: physical, datalink and network.

- The transport layer takes care of the quality and nature of data delivery. Quality of data delivery refers to issues like fast delivery versus normal delivery.

- The session layer manages how to handle packets to be transported by various networks, nodes and links. It establishes a logical session for transport.

- The presentation layer makes the data presentable (encryption/decryption, for example) for transport by syntax, technique for floating point numbers and so on.

- The application layer defines the type of application of data to be transported. It implements, for example, file system operations.

A protocol suite used in the Internet is for the standard networking of isolated networks of various types. It is made of more than one protocol. TCP/IP as well as UDP/IP are both protocol suites developed for the Internet.

TCP/IP is a four-layered protocol (Fig. 3), where layer 1 is the network interface service layer. It corresponds to layers 1 and 2 of OSI/ISO protocol. It provides services related to MAC, device drivers, physical medium, physical attachment and physical signals. In this layer, datagrams are packaged into frames.

Layer 2 is the IP layer. This layer corresponds to layer 3 of OSI/ISO protocol. It provides the routing of datagrams as units of data. Logically, these are packets. The service is connectionless; hence, switching is a datagram service rather than virtual circuit switching.

IP provides the basic service of getting the datagram to its destination. IP receives TCP packets from the upper layer TCP and then it forms its own packet, known as IP packets. Each IP packet is associated with several IP headers. For example, an original